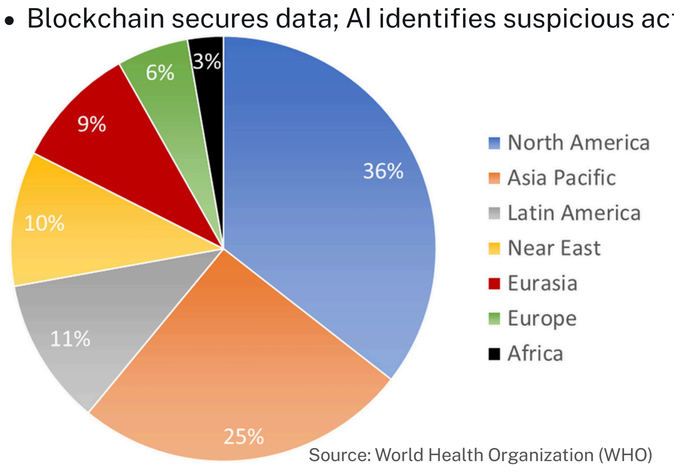
BLOCKCHAIN AND AI-DRIVEN PHARMACEUTICAL SUPPLY CHAIN FRAMEWORK FOR SECURE DRUG TRACEABILITY AND COUNTERFEIT DETECTION

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01. Introduction

- Sri Lanka's pharmaceutical supply chain (NMRA → SPC → MSD → hospitals/pharmacies) relies on fragmented, paper-based records
- **2023 immunoglobulin incident:** ~22,500 substandard vials entered state hospitals through forged documents
- Blockchain secures data; AI identifies suspicious activity.

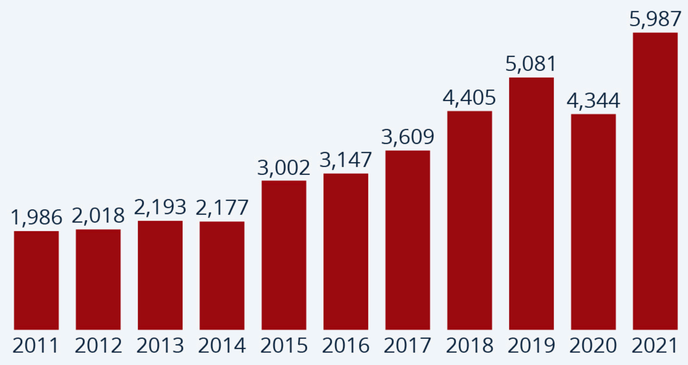


Source: World Health Organization (WHO)

Figure 1:Tackling Counterfeit Drugs

Counterfeit Drugs on the Rise Globally

Total number of counterfeit incidents concerning pharmaceuticals worldwide



Source: Pharmaceutical Security Institute

Figure 2: Counterfeit drugs Rise Globally.

02. Research Problem

Sri Lanka's pharmaceutical supply chain lacks a secure, end-to-end drug traceability mechanism.

- Records can be forged or altered without detection
- Counterfeit and substandard drugs enter legitimate channels
- Suspicious patterns detected only after harm occurs

03. Research Question

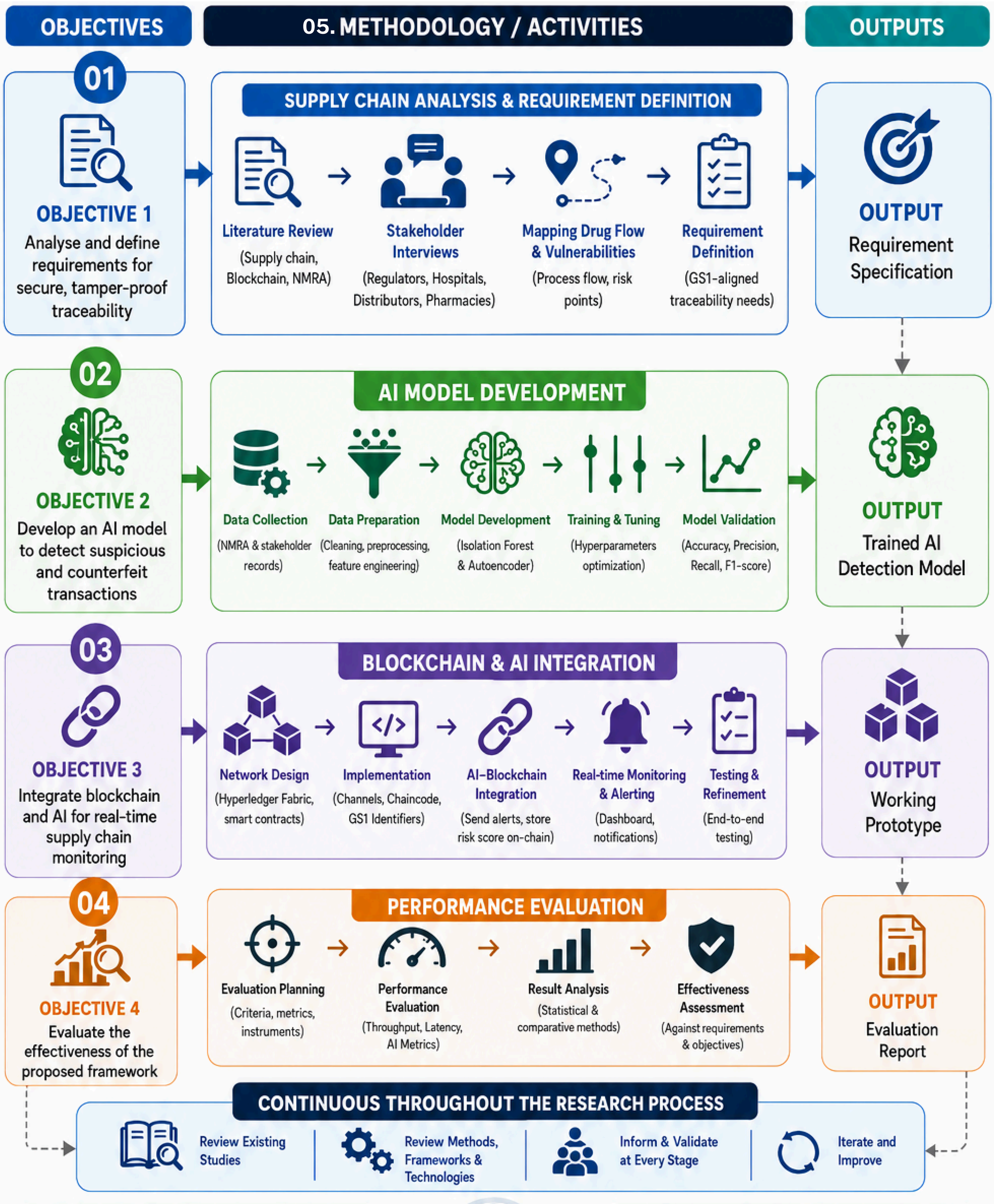
> *How can blockchain and artificial intelligence be integrated into a supply chain framework to provide secure drug traceability and effective counterfeit detection in Sri Lanka?*

04. Research Aim & Objectives

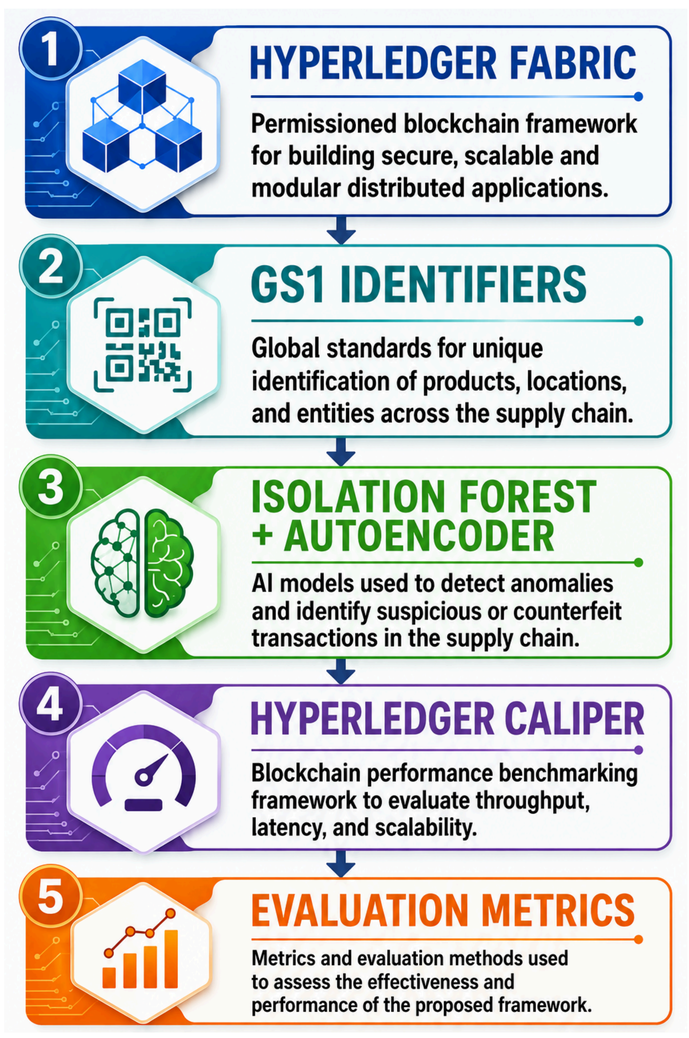
To develop and evaluate a blockchain and AI-driven pharmaceutical supply chain framework providing secure, tamper-proof drug traceability and intelligent counterfeit detection for the Sri Lankan context.

1. Analyse pharmaceutical supply chain activities for secure, tamper-proof drug traceability using blockchain
2. Develop an AI-based model to detect suspicious and counterfeit drug transactions
3. Integrate blockchain and AI for real-time supply chain monitoring
4. Evaluate the framework's effectiveness in transparency, security, and counterfeit detection

05. METHODOLOGY / ACTIVITIES



06. Key Technologies



07. References

1. World Health Organization (2017). *A study on the public health and socioeconomic impact of substandardand falsifiedmedical products*WHO, Geneva.
2. Musamih, A. et al. (2021). A blockchain-based approach for drug traceability in healthcare supply chain. *IEEE Access*, 9, 9728–9743.
3. Uddin, M. (2021). Blockchain Medledger: Hyperledger Fabric enabled drug traceability system for counterfeit drugs. *Int. J. Pharmaceutics*, 597, 120235.
4. Liu, F., Ting, K. & Zhou, Z.-H. (2008). Isolation Forest. *Proc. 8th IEEE ICDM*, 413–422.
5. Salah, K. et al. (2019). Blockchain for AI: Review and open research challenges. *IEEE Access*, 7,10127–10149.